

AP CALCULUS AB – UNIT 4

4.1 Interpretations of the Derivative



Name: _____ Class: _____ Date: _____

Lesson Overview and Learning Objectives

- Interpret a derivative value as an instantaneous rate of change in context.
- State derivative units by dividing output units by input units.
- Use the sign and magnitude of a derivative to describe local behavior.
- Distinguish the meaning of a quantity, its first derivative, and its second derivative.

Keys: Essential Knowledge and Vocabulary

Contextual derivative	If $Q = Q(t)$, then $Q'(a)$ is the instantaneous rate at which Q changes with respect to t at $t = a$.
Units	Units of $Q'(t)$ are output units per input unit; units of $Q''(t)$ are output units per input-unit squared.
Sign	$Q'(a) > 0$ means Q is increasing locally; $Q'(a) < 0$ means it is decreasing locally.
Magnitude	A larger $ Q'(a) $ means faster instantaneous change, not necessarily a larger quantity.
Complete interpretation	Name the quantity, input value, direction, numerical rate, and units.
Second derivative	$Q''(a)$ describes how the first rate Q' is changing at the instant.

Explanation, Strategy, and Common Errors

- Read the notation before calculating: identify the dependent quantity, the independent variable, and their units.
- Do not say that $Q'(a)$ is the amount of Q . It is the instantaneous rate of change of Q at input a .
- A negative derivative describes direction of change; it does not imply the original quantity is negative.
- For a table, $Q'(a)$ may be estimated using a symmetric difference when data on both sides are available.
- Interpret $Q''(a)$ as the rate of change of Q' ; include squared input units in the denominator.

Solved Example: Population interpretation

A city population $P(t)$ is measured in thousands of people, where t is years after 2025. Interpret $P'(4) = 2.7$.

Answer and Reasoning

In 2029, the city population is increasing at an instantaneous rate of 2.7 thousand people per year.

Solved Example: Marginal cost

The cost $C(q)$ is measured in dollars for producing q units. Interpret $C'(600) = 14.20$.

Answer and Reasoning

Near a production level of 600 units, total cost is increasing by approximately \$14.20 for each additional unit produced.

Solved Example: Second derivative

Temperature $T(t)$ is in degrees Celsius and time is in minutes. Interpret $T''(8) = -0.35$.

Answer and Reasoning

At minute 8, the rate at which temperature changes is decreasing by 0.35°C per minute each minute.

Solved Example: Estimate from data

A medication concentration satisfies $C(2.9) = 18.41$ and $C(3.1) = 17.73$ mg/L. Estimate and interpret $C'(3)$.

Answer and Reasoning

$C'(3) \approx \frac{17.73 - 18.41}{0.2} = -3.4$ mg/L/min. At 3 minutes, concentration is decreasing at about 3.4 mg/L each minute.

Shared Representation / Data

Quantity	Input	Derivative	Interpretation pattern
$P(t)$ people	t years	$P'(t)$ people/year	population change at time t
$C(q)$ dollars	q items	$C'(q)$ dollars/item	marginal cost near output q
$T(h)$ $^\circ\text{C}$	h hours	$T'(h)$ $^\circ\text{C}/\text{hour}$	temperature change at hour h
$V(r)$ cm^3	r cm	$V'(r)$ cm^2	sensitivity of volume to radius

AP Exam Language

For every applied derivative, include the sign, units, input value, and a complete interpretation. In related-rates work, differentiate before substituting. For L'Hôpital problems, verify an allowed indeterminate form first.

Leveled Practice**Easy Foundations**

Build fluency with the definition, notation, and one-step applications.

1. The mass $M(t)$ is in grams and t is in seconds. State the units of $M'(t)$ and $M''(t)$.

2. Interpret $V'(12) = -5.8$ when V is liters and time is minutes.

3. If $R'(4) > 0$, what does this say about R near $t = 4$?

4. If $A'(x) = 0$ at $x = 7$, what does this mean?

Medium Applications

Connect representations and apply the lesson procedure in familiar settings.

5. Revenue $R(q)$ is in dollars. Interpret $R'(120) = 38$.

6. A lake depth $D(t)$ is in meters and $D'(5) = -0.012$. Write a complete contextual statement.

7. Using $F(7.9) = 42.6$ and $F(8.1) = 43.8$, estimate $F'(8)$.

8. If $N'(10) = 5$ and $N''(10) = -0.8$, describe N and its rate at $t = 10$.

Hard Reasoning

Select a method, justify conclusions, and combine several ideas.

9. Two quantities satisfy $A'(3) = -12$ and $B'(3) = 7$. Which is changing faster at 3? Explain.

10. Explain why the statement ' $P'(5) = -2$, so $P(5) = -2$ ' is invalid.

11. A graph of f' crosses from positive to negative at $x = 2$. Describe the local behavior of f .

12. The radius of a sphere is in cm and volume in cm^3 . Explain why $V'(r)$ has units cm^2 .

Difficult Analysis

Work through less-direct algebra, subtle hypotheses, and multi-step reasoning.

13. A pollutant amount $A(t)$ satisfies $A(6) = 90$, $A'(6) = -4$, and $A''(6) = 1.2$. Give a complete local description.

14. A quantity has $Q'(a) = 0$ and $Q''(a) > 0$. What can and cannot be concluded?

15. Suppose $C'(500) = 11$ dollars/item and $C''(500) = 0.04$ dollars/item². Estimate $C'(510)$ linearly.

Challenge / AP Extension

Synthesize the topic in unfamiliar, proof-oriented, or parameter-based problems.

16. Let $Q(t) = f(g(t))$. Write a contextual interpretation of $Q'(a) = f'(g(a))g'(a)$ in words.

17. A model gives $P'(2) = 0$ but measurements show P increasing on both sides of 2. Explain how this is possible.

AP-Style Multiple-Choice Practice

Part A: No Calculator

Choose the best answer. Justification is not required on the AP multiple-choice section, but show reasoning while practicing.

1. If $H(t)$ is height in meters and t is seconds, the units of $H''(t)$ are
(A) m (B) m/s
(C) m/s^2 (D) s/m^2
2. The statement $P'(7) = -3$ means
(A) $P(7) = -3$ (B) P decreases at 3 units per input unit at 7
(C) P is negative near 7 (D) P has a minimum at 7
3. Which value indicates the fastest change?
(A) $f'(a) = -9$ (B) $g'(a) = 6$
(C) $h'(a) = 0.5$ (D) $k'(a) = -4$
4. If $T'(4) > 0$ and $T''(4) < 0$, then at $t = 4$ the temperature is
(A) decreasing at an increasing rate (B) increasing at a decreasing rate
(C) at a local minimum necessarily (D) constant

Part B: Graphing Calculator Permitted

Use a calculator only for numerical evaluation, graphing, or regression as indicated. Preserve mathematical setup.

5. A calculator table gives $P(4.99) = 81.72$ and $P(5.01) = 81.96$. A symmetric estimate of $P'(5)$ is
(A) 0.12 (B) 1.2
(C) 12 (D) 24
6. For $C(q) = 1500 + 9q + 0.002q^2$, the marginal cost at $q = 400$ is
(A) 9.8 (B) 10.6
(C) 11.4 (D) 1510.6

AP-Style Free-Response Practice

Calculator Permitted

A reservoir volume $V(t)$, in millions of liters, is measured t days after a storm.

(a) Given $V(3) = 48.2$ and $V'(3) = -1.7$, interpret both values. [2 points]

(b) Given $V''(3) = 0.25$, interpret this value. [2 points]

(c) Use a linear approximation to estimate $V(3.4)$. [2 points]

No Calculator

The number of users of an application is $N(t) = 2000 + 600t - 40t^2$, where t is months after launch.

(a) Find $N'(t)$ and state its units. [2 points]

(b) Interpret $N'(5)$. [2 points]

(c) Find and interpret $N''(5)$. [2 points]